

Automatic Generation of Model and Data Cards: A Step Towards Responsible AI

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Abstract

In an era of model and data proliferation in machine learning/AI especially marked by the rapid advancement of open-sourced technologies, there arises a critical need for standardized consistent documentation. Our work addresses the information incompleteness in current human-generated model and data cards. We propose an automated generation approach using Large Language Models (LLMs). Our key contributions include the establishment of CARDBENCH, a comprehensive dataset aggregated from over 4.8k model cards and 1.4k data cards, coupled with the development of the CARDGEN pipeline comprising a two-step retrieval process. Our approach exhibits enhanced completeness, objectivity, and faithfulness in generated model and data cards, a significant step in responsible AI documentation practices ensuring better accountability and traceability.¹

1 Introduction

The landscape of artificial intelligence (AI) has undergone a profound transformation with the recent surge in open-sourced models (Villalobos et al., 2022; Sevilla et al., 2022) and datasets (Northcutt et al., 2021; Sevilla et al., 2022). The trend has been significantly accelerated by the advent of disruptive technologies such as transformers (Gruetzmacher and Whittlestone, 2022; Vaswani et al., 2017). Since this proliferation of accessible models and datasets can have their applications significantly influence various aspects of society, it becomes increasingly important to underscore the necessity for standardized consistent documentation to communicate their performance characteristics accurately (Liang et al., 2022).

In this context, model cards proposed by Mitchell et al. (2019) and data cards proposed by Pushkarna

¹Our code and data is available at <https://github.com/jiarui-liu/AutomatedModelCardGeneration>.

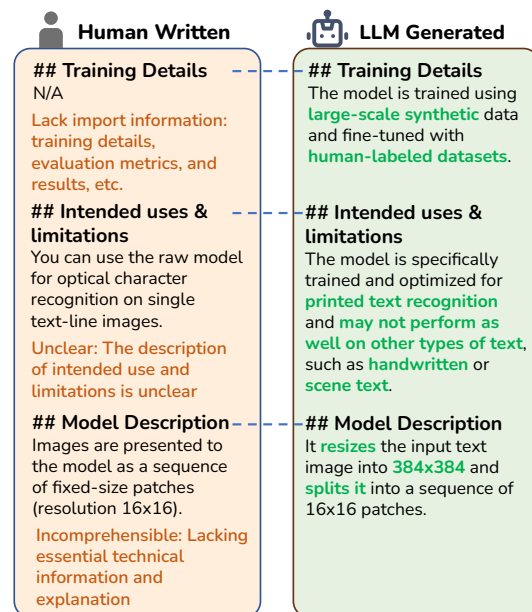


Figure 1: Common problems with manually generated model cards and data cards.

et al. (2022), emerge as necessary documentation tools. These cards bridge the communication gap between model/data creators and product developers, thereby ensuring a comprehensive understanding of the model’s/data’s capabilities and limitations in both academic and industrial applications (Pushkarna et al., 2022; Sevilla et al., 2022; Vaswani et al., 2017; Sevilla et al., 2022). Model/data cards are instrumental in research, offering detailed insights such as data characteristics, sources, etc, as well as model architecture, training procedures, and potential biases and limitations, which accelerates development and reduces error propagation in subsequent models (Swayamdipta et al., 2020).

Inspired by these concepts, HuggingFace (HF) developed card specifications for models and datasets hosted on its website. Despite the release of some

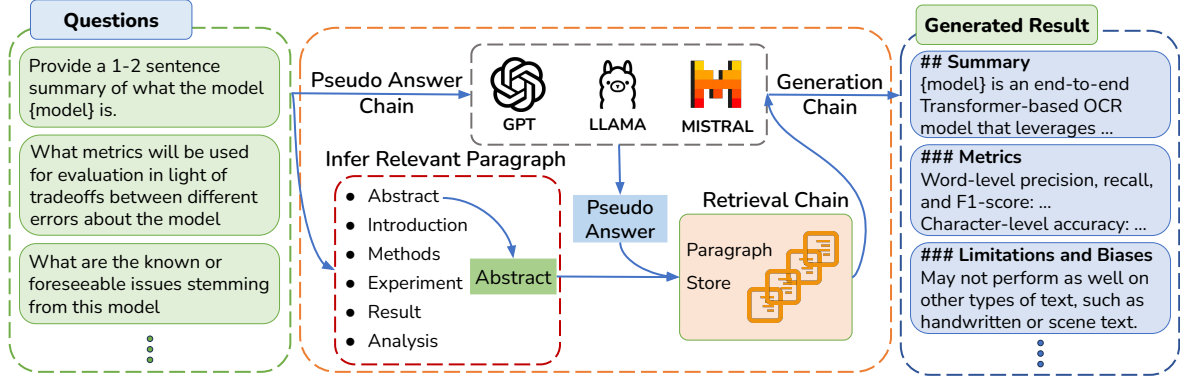


Figure 2: Overview of the CARDGEN pipeline to generate a full model card or a full data card.

available tools to assist model card writing², HF leaves the decision of what to report up to developers. This raises several problems: First, this approach relies heavily on the developers’ understanding and interpretation of what should be reported, leading to inconsistencies and potential omissions of critical information (Shukla et al., 2021). Second, there is a tendency among card creators to use existing cards as templates rather than starting from the standardized template provided (Pushkarna et al., 2022). Such variability compromises the comprehensiveness and reliability of the cards.

With the power of state-of-the-art LLMs (Touvron et al., 2023; Brown et al., 2020; Ouyang et al., 2022; Jiang et al., 2023; Touvron et al., 2023), the automatic generation of model and data cards presents a method to ensure uniformity, consistency, and thoroughness across various model/dataset documentation. To this end, we contribute the following: (1) A novel pioneering initiative to systematically utilize LLMs for automatically generating model/data cards; (2) CARDBENCH, a curated dataset that encompasses all the associated papers and GitHub READMEs referenced in 4.8k model cards and 1.4k data cards; (3) A novel approach that decomposes the card generation task into multiple sub-tasks, proposing a CARDGEN pipeline including a two-step retrieval process; (4) A novel set of quantitative and qualitative evaluation metrics. We demonstrate that using our pipeline with GPT3.5, we achieve higher scores than human generated cards on completeness, objectivity, and understandability, demonstrating the effectiveness of the CARDGEN pipeline.

²https://huggingface.co/spaces/huggingface/Model_Cards_Writing_Tool

2 Related Work

2.1 Accountability and Traceability for AI Systems Through Documentation

The increasing complexity of AI systems has raised significant concerns regarding their potential biases and lack of transparency, which in turn poses negative implications for users and society (Jacovi et al., 2021; Barocas and Selbst, 2016; Panch et al., 2019; Daneshjou et al., 2021; Huang et al., 2023). This has motivated the emergence of various documentation frameworks for ML models and datasets:

Model Cards Mitchell et al. (2019) introduced the concept of model cards as a framework for the transparent documentation of machine learning (ML) models and provided detailed evaluations across diverse demographic groups and conditions. Subsequent advancements in model card design have included advocating for the generation of consumer labels for ML models (Seifert et al., 2019), introducing principles for explainable models (Phillips et al., 2020), suggesting other cards as complements to model cards (Adkins et al., 2022; Shen et al., 2021), environmental and financial impact considerations (Strubell et al., 2019), and some toolkits that help to track and report specific information in ML models (Arya et al., 2019; Shukla et al., 2021).

Data Cards In the domain of ML dataset documentation, Gebru et al. (2021) pioneered the concept of datasheets for datasets, followed by the introduction of data statements for NLP data (Bender and Friedman, 2018; Bender et al., 2021), and the concept of data nutrition labels to aid in better decision-making (Holland et al., 2020).

McMillan-Major et al. (2021); Hutchinson et al. (2021) provided comprehensive data card templates. Pushkarna et al. (2022) proposed data cards for responsible AI development. Díaz et al. (2022) introduced CrowdWorkSheets for the transparent documentation of crowdsourced data. Our work builds upon the existing model and data card documentation templates released by HF.

2.2 Knowledge-Enhanced Text Generation

LLMs can be augmented with external knowledge sources to improve their reasoning capabilities (Lewis et al., 2020; Li et al., 2022). Retriever, generator, and evaluator are the key components in a standard RAG system. With the advancement of powerful pretrained seq2seq models as generators, numerous studies have concentrated on the evaluation performance:

RAG Text Generation Evaluation Due to variations in retrieved content, customized generation pipelines, and user intentions, evaluating the effectiveness of LLM generated texts in a Retrieval-Augmented Generation (RAG) system becomes challenging (Huang et al., 2023; Mialon et al., 2023). Traditional n -gram based metrics like BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), and PARENT-T (Wang et al., 2020b) are used for assessing the overlap between generated texts and references, but cannot fully grasp the quality nuances of human expectations (Honovich et al., 2021; Maynez et al., 2020). Some model-based metrics have later been invented to better align with human judgments without requiring supervision, such as BERTScore (Zhang et al., 2019), MoverScore (Zhao et al., 2019), and BARTScore (Yuan et al., 2021). Research has primarily focused on factuality (Gou et al., 2023; Chen et al., 2023; Galitsky, 2023; Min et al., 2023), and faithfulness (Barrantes et al., 2020; Fabbri et al., 2022; Santhanam et al., 2021; Laban et al., 2023; Durmus et al., 2020) of generated content. Some frameworks have been designed to automate the assessment pipeline utilizing the capabilities of LLMs (Es et al., 2023; Pietsch et al., 2020; Liu et al., 2023; Fu et al., 2023; Manakul et al., 2023). In this work, we present a comprehensive evaluation of our approach using both traditional metrics and LLM-based automatic metrics. Additionally, we offer a detailed human evaluation of multiple performance aspects, including faithfulness.

3 Defining the Model/Data Card Generation Task

3.1 Task Formulation

Denote our test set as $D := \{(m_i, p_i, g_i)\}_{i=1}^N$ consisting of N triples, each with a human-generated model card m_i , a direct paper document p_i , and a direct GitHub README document g_i . For each question q_j from the question template set $Q := \{q_j\}_{j=1}^M$, we define a two-stage retrieve-and-generate task f_1 and f_2 .

The retrieval task $f_1 : \mathcal{P} \times \mathcal{G} \times \mathcal{Q} \rightarrow \mathcal{R}$ maps source paper and GitHub documents according to the question to a set of retrieved chunks R .

The generation task $f_2 : \mathcal{R} \times \mathcal{Q} \rightarrow \mathcal{A}$ maps the retrieved chunk set and questions to a space $\mathcal{A}$ that contains generated answers for all questions.

3.2 Structured Generation

Inspired by the model card design from Mitchell et al. (2019), HF provides its guidelines about how to fully fill out a model card.³ It suggests a detailed disclosure of the model features and limitations in a published model card. Following the guidelines, we define seven sections including 31 individual questions for generating a complete model card. These sections are model summary, model details, uses, bias and risks, training details, evaluation, and additional information about the proposed model. We have made our full question template for both model cards and data cards accessible in Appendix A. Table 1 highlights the most important questions for each section of the full template.

4 CARDBENCH Dataset

CARBENCH contains 4,829 human-generated model cards and 328 data cards with paper and GitHub references.

4.1 Dataset Collection

Data Source and Preprocessing We identify the model page⁴ and the dataset page⁵ on HF as data sources. We crawl the model cards and data cards (READMEs) associated with the 10,000 most downloaded models and datasets, respectively, from the HF page as of October 1, 2023. For each collected model card, we use regular expressions

³<https://huggingface.co/docs/hub/model-card-annotated>

⁴<https://huggingface.co/models>

⁵<https://huggingface.co/datasets>

Question	Role	Prompt
Summary	Project organizer	Provide a 1-2 sentence summary of what the model is.
Description	Project organizer	Provide basic details about the model. This includes the model architecture, training procedures, parameters, and important disclaimers.
Direct use	Project organizer	Explain how the model can be used without fine-tuning, post-processing, or plugging into a pipeline. Provide a code snippet if necessary.
Bias, risks, limitations	Practical Ethicist	What are the known or foreseeable issues stemming from this model? These include foreseeable harms, misunderstandings, and technical and sociotechnical limitations.
Results summary	Developer	Summarize the model evaluation results.

Table 1: Template of the most important questions for each section. “Roles” are provided as role specifications in Appendix Figure 8 for LLMs, and “prompts” are provided as queries.

to find all valid paper URLs and GitHub repository URLs. We leverage the SciPDF Parser⁶ to parse downloaded paper PDFs into a JSON formatted data structure, capturing the paper sections. We further use the GitHub REST API⁷ to obtain README files from each repository. For each collected data card, we devise regular expressions to locate all data cards with the “Dataset Description” section, which should contain information such as the dataset homepage, paper link, and GitHub repository. Then, based on the information obtained from the data card, we retrieve and process paper documents and GitHub READMEs as done for model cards.

Evaluation Set Construction In the absence of standardized and strict content requirements by HF, collected model cards are mostly incomplete, and some examples are even minimally modified copies of existing ones. This variability undermines the reliability of our comparative evaluation against human-generated model cards as a reference metric. In an attempt to mitigate this shortcoming, we curate the highest quality human generated model cards to serve as our evaluation data set. This set comprised a select 350 examples that are rewritten by the HF team with their unique disclaimers. Also, for data cards, the majority of those collected are incomplete and lack content readability. In order to have a sufficient number of evaluation sets, we first selected all the data cards with a “Dataset Description” section. We then wrote markdown matching logic to obtain 300 examples as our evaluation set based on the word count and the number of sections in the data cards. See Appendix B for more details on data collection.

⁶https://github.com/titipata/scipdf_parser

⁷<https://docs.github.com/en/rest?api>

	Split	Paper		GitHub	
		# Sections	# Words	# Sections	# Words
ModelCard	all	29	6810	22	2495
	test	30	6674	17	1855
DataCard	all	25	5741	9	975
	test	25	5784	8	816

Table 2: Statistics for direct paper documents and repository READMEs for crawled model cards and data cards, in terms of the average number of sections and the average number of words of documents.

4.2 Data Annotation

In our methodology for generating model cards, we predominantly focus on the model’s design detail itself rather than referencing external methodologies cited in human-generated model cards. It necessitates the identification of the primary paper proposing the model, along with the direct repository reflecting model implementation. The evaluation set is annotated by two ML Master’s student researchers who know HF models well and are proficient in English. The process resulted in 294 evaluation examples having both direct paper and repository links. Additionally, to annotate the whole dataset, we prompt GPT-3.5-Turbo (Brown et al., 2020) to validate direct source document links, given the context wherein each URL is situated in the model card. We finally obtained 4,829 non-empty ones with either direct paper links or repository links. GPT’s annotation reached 98.01% accuracy according to human validation results on the test set. For data cards, their primary paper link and direct repository responsible for the dataset is within the ‘Dataset Description’ section. We finally obtained 865 data cards with either direct paper links or repository links. This gain resulted in 99.7% accuracy according to human validation results on the 300 data cards test set. See Appendix C for human annotation guidelines and prompts for GPT validation.

4.3 Data Statistics

We show the overall statistics in Table 2 and Appendix Table 13. We can observe that our test set, the set of model cards rewritten by the HF team, are more concise than other developer-written ones. Their corresponding source documents have similar sizes in terms of the number of sections and words.

To explore whether our test set represents the whole dataset well, we look into some model card features obtained with the HF API. Appendix Figure 6 shows that test set examples are nearly uniformly distributed compared to the overall dataset in terms of the number of downloads, and task distributions of models/datasets. A comparison of the test set to the whole set is shown in Appendix Figure 5. See Appendix D for additional dataset analyses.

5 Method: the CARDGEN Pipeline

5.1 Overview

Figure 2 shows our CARDGEN pipeline. For each q_j in Q , we first prompt LLMs to split q_j into a sub-question set. Next, we use LLMs to infer relevant sections as potential knowledge sources, and generate pseudo answers for each sub-question leveraging LLM’s own knowledge (Gao et al., 2023). The pseudo answer is used as a query to get the set R of relevant document chunks. We use an LLM to generate answers for the question prepended with highest-ranked document chunks.

5.2 Designing the Retriever

As the process of supervised retrieval necessitates the acquisition of additional crowd-sourced annotations for establishing ground truth sentences for each query, it constitutes a substantial amount of labor. Consequently, we choose to modify the standard RAG retrieval baselines (Lewis et al., 2020), where source documents are ranked based on the inner product similarity with a query question. We develop a two-step retrieval method to improve the retrieval precision: (1) Given all section names of a model’s paper and README documents, we prompt the LLM to infer the top-k most plausibly relevant sections. (2) We query the pseudo answer from chunks in the inferred section contents after feeding it into an embedding model. We use the embedding model jina-embeddings-v2-base-en developed by Günther et al. (2023). This choice is further verified in Sections 8.2 and 8.3.

5.3 Designing the Generator

For our CARDGEN pipeline, we test Claude 3 Opus (Anthropic, 2024), GPT-4-Turbo (OpenAI, 2023), GPT-3.5-Turbo (Brown et al., 2020), Llama2 70B Chat (Touvron et al., 2023), Vicuna 13B V1.5 (Zheng et al., 2024), Llama2 7B Chat (Touvron et al., 2023), Mistral 7B Instruct (Jiang et al., 2023) as backbone LLMs. We generate an answer t_j for each question q_j based on R , and concatenate all answers in sequence to form the final model card. To leverage the LLM’s strengths in effectively responding to varied questions, we assign specific roles to the LLM tailored to different questions, and outline its expected areas of expertise. The pre-defined roles, such as project organizer, sociotechnical practical ethicist, and developer, are outlined in Table 1 and Appendix A, as noted by Raw et al. (2022). See Appendix F for LLM inference details.

6 Baselines

We evaluate our two-step retrieval and generation processes in CARDGEN against two baselines:

(1) **One-step retrieval:** By keeping all other components of our pipeline unchanged, we reduce the current two-step retrieval method to a one-step pipeline by directly retrieving the top-12 chunks from the entire paper and GitHub documents without first inferring relevant paragraphs. Although intuitively the nature of our question template set correlates closely with the sectional structure of papers and GitHub repositories, this baseline could provide further support of using a paragraph-level retriever.

(2) **Retrieval only:** Upon completing the two-step retrieval and obtaining relevant chunks, the method directly use the retrieved chunk as the final output. This is used to assess the advantages of the summary generation step over merely using the author’s original text.

We compare our CARDGEN pipeline with these two baselines in Section 8.2.

7 Evaluation Setup

We evaluate CARDGEN on various standard as well as state-of-the-art metrics to measure the faithfulness, relevance, and other aspects of the generation quality. Additionally, we incorporate human evaluation for the pipeline to address three key chal-

Metric	Input	Description
Factual consistency	R, A	How much the generated answer is supported by retrieved contexts.
Faithfulness	Q, R, A	How much the statements created from the question-answer pair are supported by the retrieved context.
Answer relevance	Q, A	relevance score of the answer according to the given question.
Context precision	Q, R	How much the given context is useful in answering the question.
Context relevance	Q, R	Whether the question can be answered by relevant sentences extracted from the given context.

Table 3: Illustration of the input, along with a description of standard metrics and GPT-based metrics being used. Here Q, R, A represent the questions, retrieved texts, and generated texts, respectively.

Metric	Human	Claude3 Opus	GPT4	GPT3.5	Llama2 70B	Vicuna 13B	Llama2 7B	Mistral 7B
Completeness	1.92	7.28	5.99	5.24	4.76	4.24	2.50	4.07
Accuracy	6.66	6.56	6.04	4.51	3.61	3.11	1.84	3.67
Objectivity	2.03	7.16	6.33	5.23	4.72	4.25	2.12	4.16
Understandability	2.49	7.11	6.21	4.99	4.80	3.80	2.09	4.51
Reference quality	6.13	6.75	5.40	4.28	4.15	3.73	1.63	3.93

Table 4: Human evaluation results on LLM generated and human-generated model cards.

lenges that can’t be solved by automatic metrics alone: First, there is an absence of ground truth labels of generated model cards by CARDGEN. To mitigate this, we have to develop specific manual evaluations to assess performance. Second, current model cards created by human developers are often incomplete and deviate from the recommended template provided by HF. Third, the LLM generated model card is typically long with over 4000 words, and brings challenges to both open-source standard evaluations with limited context size and costly GPT-based metrics.

Standard Metrics We follow Honovich et al. (2022) and use ROUGE (Lin, 2004), BERTScore (Zhang et al., 2019), BARTScore (Yuan et al., 2021), and NLI-finetuned models (Williams et al., 2018; MacCartney and Manning, 2008) to measure the factual consistency of retrieved chunks set R and the generated answer A . Due to the large size of retrieved texts, we use deberta-v3-base as the base model for BERTScore, and use nli-deberta-v3-large as the NLI-finetuned model scorer (Reimers and Gurevych, 2019a; He et al., 2021). More details in Appendix H.

GPT Metrics Following Es et al. (2023), we consider the measurement of faithfulness, answer relevance, context precision, and context relevance using GPT4. Table 3 provides a description of these metrics. As different combinations of inputs are taken into consideration, these metrics are necessary supplements to standard metrics. Full prompt details are explained in Appendix H.

Human Evaluation Metrics Putting together LLM generated cards with the human-generated cards as a sample, we devise the following manual evaluation metrics: completeness, accuracy, objectivity, understandability, and reference quality. We design a simple Gradio annotation interface (Abid et al., 2019), and more details are in Appendix I.

8 Results

8.1 Performance Summary

Our human evaluation results are shown in Table 4 and automatic evaluation results are shown in Tables 5 and 6 for model cards. The only difference for the data card generation pipeline is the substitution with data card question templates. In this subsection, we mainly answer two questions below:

Are our generated model cards better than human-generated ones? We conduct a random sampling of 50 model cards from the test set and compute the average metric scores across all the annotated samples, as shown in Table 4. GPT3.5, GPT4, and Claude3 Opus demonstrates superior performance over other open-sourced LLMs and human-generated content in terms of completeness, objectivity, and understandability. This finding aligns with the observations presented below for Tables 5 and 6.

Conversely, the human-generated model cards often received higher scores in accuracy and reference quality. This disparity suggests that all LLMs exhibit some degree of hallucination for factual content and reference links in their generation. It is

Metric	Model	Summary	Model details	Uses	Bias	Training details	Evaluation	More info	All
ROUGE-L	Claude3 Opus	8.91	11.26	14.39	14.21	14.11	14.99	12.78	13.04
	GPT4	8.80	9.35	15.38	18.20	17.59	19.40	9.73	13.27
	GPT3.5	9.90	10.70	16.51	20.21	14.46	15.75	10.73	13.16
	Llama2 70b chat	12.71	14.35	12.85	17.20	18.74	18.03	16.21	15.98
	Vicuna 13b v1.5	10.78	11.35	13.54	17.10	16.06	16.75	10.29	13.12
	Llama2 7b chat	11.91	12.84	13.89	15.85	14.63	16.21	13.61	14.08
	Mistral 7b inst	12.19	11.01	13.02	15.07	16.79	16.23	9.47	12.70
BERTScore	Claude3 Opus	54.78	53.73	58.42	56.32	57.83	58.80	55.17	56.10
	GPT4	54.06	50.44	57.81	58.81	59.50	61.24	47.48	53.96
	GPT3.5	54.86	53.17	58.62	59.29	56.61	57.42	52.47	55.09
	Llama2 70b chat	57.21	56.15	53.97	56.55	59.69	59.46	56.99	57.21
	Vicuna 13b v1.5	55.15	52.97	54.99	57.24	57.61	58.83	52.10	54.83
	Llama2 7b chat	55.76	54.51	53.93	55.48	56.30	57.13	54.72	55.26
	Mistral 7b inst	55.69	52.80	54.12	53.76	57.10	57.63	49.12	53.47
BARTScore	Claude3 Opus	13.92	5.60	2.56	1.59	4.10	2.87	4.33	4.36
	GPT4	9.69	7.63	1.43	1.98	4.02	4.29	6.11	5.34
	GPT3.5	17.09	9.58	2.04	3.52	5.75	6.65	9.10	7.61
	Llama2 70b chat	14.17	5.41	1.45	3.10	5.30	4.60	5.91	5.15
	Vicuna 13b v1.5	13.53	5.67	1.90	3.76	5.63	6.81	6.77	5.90
	Llama2 7b chat	14.04	3.49	2.11	3.61	4.70	3.68	4.01	4.03
	Mistral 7b inst	16.52	9.65	2.00	3.55	7.00	8.75	8.31	7.90
NLI	Claude3 Opus	58.00	54.62	56.33	59.00	62.25	61.40	60.12	58.68
	GPT4	61.00	52.88	53.00	56.00	64.50	65.60	62.62	59.42
	GPT3.5	65.14	49.83	57.54	62.41	59.14	60.14	56.80	56.54
	Llama2 70b chat	56.46	51.70	55.22	58.42	57.70	62.04	59.74	57.14
	Vicuna 13b v1.5	60.20	51.40	58.05	55.10	58.29	63.33	55.00	56.31
	Llama2 7b chat	56.46	50.19	54.31	57.23	57.82	62.11	56.44	55.77
	Mistral 7b inst	58.67	50.36	54.25	54.59	59.06	58.91	55.17	55.02

Table 5: Factual consistency evaluation results per section on our retrieve-and-generate pipeline using ROUGE-L, BERTScore, BARTScore, and NLI pretrained scorers.

important to note that the human-generated model cards’ incompleteness precludes a direct comparison of human evaluation metrics with the metrics used in Tables 5 and 6. Moreover, the insights derived from Table 4 are not obtainable through automatic metrics. We thus conclude that human evaluation metrics are indispensable components of our overall evaluation framework.

How does GPT3.5 perform compared with open sourced LLMs? From Table 5, we can’t observe a uniform trend for factual consistency across all sub-tasks. GPT3.5 outperforms open-sourced LLMs on “Uses” and “Bias” question sets in 3 over 4 standard metrics, while Llama2 70b generates more factual consistent answers on other sub-tasks according to ROUGE-L and BERTScore.

According to Table 6, GPT3.5 beats other LLMs on faithfulness and answer relevance across nearly all sub-tasks, and shows its strong instruction-following capabilities for question-answering. However, we have an interesting observation that though GPT3.5 has higher context relevance scores, it is outperformed by Mistral 7B on context precision. A higher context relevance indicates that

the question can be better answered from the given context, while a lower context precision means that the context may contain other unnecessary information for answering the question. The discrepancy between results by these two metrics suggests that retrieved texts from the GPT CARDGEN pipeline are more informative but less concise. Additionally, since we use LLM generated pseudo answers as queries for similar paragraphs, pseudo answers with more possibly unrelated contents will lead to more irrelevant chunks from retrieval. Along with the illustration in Appendix Figure 7, we draw the conclusion that GPT3.5 generates pseudo answers with potentially more unrelated details.

8.2 Baseline Results

To assess the effectiveness of CARDGEN’s retriever and generator, we first compare it to the baseline methods outlined in Section 6. To manage the expenses associated with OpenAI AI calling, we employ GPT3.5 for subsequent studies. We obtain Krippendorff’s α (mean=0.83, std=0.14, min=0.56, max=0.99) for the agreements on Table 6 by GPT4 and GPT3.5 to validate our evaluation model substitution (Castro, 2017).

Metric	Model	Summary	Description	Direct use	Bias, risks, limitation	Results summary
Faithfulness	Claude3 Opus	74.97	49.77	78.23	71.28	84.89
	GPT4	68.87	85.58	62.99	64.20	86.44
	GPT3.5	71.23	83.21	48.71	55.17	82.99
	Llama2 70b chat	70.03	76.39	43.20	32.14	63.87
	Vicuna 13b v1.5	78.46	81.74	45.94	46.64	78.22
	Llama2 7b chat	72.41	71.35	48.43	44.23	65.56
	Mistral 7b inst	76.75	75.03	38.28	41.77	73.61
Answer relevance	Claude3 Opus	90.42	91.10	89.12	91.39	93.15
	GPT4	90.83	93.12	89.69	92.03	91.36
	GPT3.5	91.18	93.26	90.70	93.75	93.24
	Llama2 70b chat	90.76	92.27	91.25	92.23	91.63
	Vicuna 13b v1.5	89.00	91.22	90.17	92.99	90.38
	Llama2 7b chat	90.44	90.95	92.55	92.69	92.81
	Mistral 7b inst	90.46	91.77	90.36	91.56	90.43
Context precision	Claude3 Opus	33.25	51.73	26.17	20.99	42.58
	GPT4	35.01	51.25	29.29	22.76	40.23
	GPT3.5	29.07	51.80	25.71	18.77	37.88
	Llama2 70b chat	21.05	50.00	25.35	20.03	40.82
	Vicuna 13b v1.5	24.91	51.22	24.00	8.93	39.00
	Llama2 7b chat	32.46	50.79	25.52	14.27	40.04
	Mistral 7b inst	31.10	52.22	28.45	21.36	44.45
Context relevance	Claude3 Opus	13.32	48.82	28.90	21.32	23.01
	GPT4	12.86	52.39	26.63	18.89	23.47
	GPT3.5	13.27	51.03	29.82	18.97	26.44
	Llama2 70b chat	13.32	49.62	27.22	18.37	24.31
	Vicuna 13b v1.5	13.83	51.32	27.00	14.03	23.08
	Llama2 7b chat	13.87	50.78	28.07	17.57	26.23
	Mistral 7b inst	13.22	47.05	28.40	18.75	23.52

Table 6: GPT4 evaluation results on five most important questions based on faithfulness (Faith), answer relevance (AR), context precision (CP), and context relevance (CR).

Model	Method	CP	CR
GPT3.5	One-step retrieval	44.03	27.82
	CARDGEN	44.67(+0.64)	28.24(+0.42)
Llama2 70B	One-step retrieval	42.94	28.10
	CARDGEN	44.03(+1.09)	28.83(+0.73)
Llama2 7B	One-step retrieval	43.47	27.35
	CARDGEN	41.91(-1.56)	28.00(+0.65)
Mistral 7B	One-step retrieval	43.75	27.80
	CARDGEN	45.24(+1.49)	27.97(+0.17)

Table 7: GPT3.5 evaluation results of the one-step retrieval baseline and CARDGEN in terms of context precision and context relevance.

One-step retrieval Since the change is only in the retrieval process in comparison to CARDGEN, we focus exclusively on context precision and context relevance as metrics. These metrics evaluate the quality of the retrieved text r_i in response to a given question q_i . We evaluate across four LLMs, and report results based on the averaged score of the most important questions. According to Table 7, the two-step retrieval process achieves marginally yet consistently higher scores than the one-step retrieval across nearly all models. These findings indicate that a paragraph-level retrieval model constitutes a more appropriate method for this study.

Model	Method	AR	Understandability
GPT3.5	Retrieval only	81.28	5.60%
	CARDGEN	90.84(+9.56)	94.40%
Llama2 70B	Retrieval only	81.61	1.60%
	CARDGEN	90.32(+8.71)	98.40%
Llama2 7B	Retrieval only	81.32	4.40%
	CARDGEN	90.78(+9.46)	95.60%
Mistral 7B	Retrieval only	81.49	2.40%
	CARDGEN	89.83(+8.34)	97.60%

Table 8: GPT3.5 evaluation results of the retrieval-only baseline and CARDGEN in terms of answer relevance and understandability. Full results including assessments of brevity can be found in Appendix Table 15.

Retrieval only Following the same evaluation setup as above, we consider answer relevance of generated text g_i according to a given question q_i . To further compare which method produces more understandable and concise outputs, we also incorporate understandability and brevity into our evaluation as GPT metrics for pairwise comparison (Liusie et al., 2024; Fu et al., 2023). As illustrated in Table 8, CARDGEN significantly outperforms the retrieval-only baseline across all metrics, highlighting the importance of the generation step in summarizing and restating sentences from source

Metric	Model	Summary	Description	Direct use	Bias, risks, limitation	Results summary
NLI	GPT3.5	65.14(+2.14)	51.53(+0.53)	50.51(+0.51)	64.12(+1.12)	58.50(+0.50)
	w/o pseudo	63.00	51.00	50.00	63.00	58.00
Faith	GPT3.5	81.93(+6.75)	79.30(+4.30)	41.23(+0.62)	46.42(-2.53)	72.66(+1.21)
	w/o pseudo	75.18	75.00	40.61	48.95	71.45
AR	GPT3.5	86.94(+0.06)	89.56(-0.65)	88.95(+0.78)	93.55(+0.40)	95.20(+0.02)
	w/o pseudo	86.88	90.21	88.17	93.15	95.18
CP	GPT3.5	47.53(+7.49)	19.61(+1.01)	13.44(+3.20)	13.03(-0.26)	64.15(+0.24)
	w/o pseudo	40.04	18.60	10.24	13.29	63.91
CR	GPT3.5	11.85(+2.32)	23.24(-2.21)	8.70(+1.19)	4.35(+0.69)	24.04(+5.79)
	w/o pseudo	9.53	25.45	7.51	3.66	18.25
Faith	GPT3.5	81.93(+8.09)	79.30(+15.31)	41.23(+26.62)	46.42(+22.14)	72.66(+25.16)
	Llama2 70B	73.84	63.99	14.61	24.28	47.50
AR	GPT3.5	86.94(-1.56)	89.56(+0.63)	88.95(+6.58)	93.55(+9.53)	95.20(+7.21)
	Llama2 70B	88.50	88.93	82.37	84.02	87.99

Table 9: GPT3.5 evaluation results on five most important questions for pseudo answer chain ablation in top five rows and generation chain ablation in bottom two rows. For the generation chain ablation, we keep all previous chains unchanged with GPT-3.5-turbo as the backbone, and only vary the choice of LLMs for the final generation chain, including GPT-3.5-turbo and Llama2-70B-Chat-HF.

documents to enhance their understandability and conciseness. Further details are in Appendix J.

8.3 Ablation Study

We also conducted the following ablation studies and explored model architecture variations to further validate CARDGEN’s components: (1) Remove the pseudo answer chain and use original questions for embedding similarity matching. (2) Vary the final generation chain only with different LLMs, and maintain all preceding reasoning chains as generated by GPT3.5. (3) Employ different embedding models for dense retrieval.

Pseudo Answer Chain We compare the GPT evaluation scores and factual consistency using NLI of CARDGEN + GPT3.5 pipeline with or without the pseudo answer chain, as illustrated in Table 9. CARDGEN with the pseudo answer chain outperforms the other across nearly all important questions and metrics being tested. Our results demonstrate the necessity of the pseudo answer chain in our pipeline. Some lower scores may be because of more unrelated texts from the generated pseudo answers for specific questions.

Generation Chain In bottom two rows of Table 9, we show the comparison results by only substituting GPT3.5 in the generation chain with Llama2 70B based on faithfulness and answer relevance. Context precision and context relevance are the same since retrieved texts remain unchanged. We observe a large drop for the faithfulness score

and a moderate drop for the answer relevance score, indicating the stronger instruction following capability of GPT3.5 in the generation stage compared to Llama2 70B.

Embedding Models We compare the embedding model jina-embeddings-v2-base-en that we use with two other commonly used sentence transformer models: all-MiniLM-L6-v2 and all-mpnet-base-v2 (Günther et al., 2023; Wang et al., 2020a; Reimers and Gurevych, 2019b, 2020). We justify our choice of embedding models in Appendix Figure 12, where CARDGEN with jina-embeddings-v2-base-en performs better than others according to all three metrics related to the retrieved texts.

8.4 LLM Generated Model Card Statistics

Appendix G provides related statistics. Compared with statistics in Table 13, LLM generates longer and more informative than human.

9 Conclusion

In this study, we introduce a novel task focused on the automatic generation of model cards and data cards. This task is facilitated by the creation of the CARDBENCH dataset, and the development of the CARDGEN pipeline leveraging state-of-the-art LLMs. The system is designed to assist in the generation of understandable, comprehensive, and consistent models and data cards, thereby providing a valuable contribution to the field of responsible AI.

Limitations

One limitation of our method is that, despite the adoption of the RAG pipeline and explicit instructions for LLMs to adhere closely to the retrieved text, there remains the potential for hallucinations in the generated text. To mitigate this, future work may integrate specific strategies into our CARDGEN pipeline for hallucination reduction by carefully balancing generation speed with quality.

Our current approach employs a single-step generation process and a two-step retrieval process that first infers relevant section contents. Future work could incorporate more advanced chain-of-thought prompting techniques and compare with our CARDGEN pipeline. For complex questions requiring multistep reasoning, after decomposed into manageable sub-questions, we can address each sub-question through multiple reasoning steps, as suggested by recent research (Yao et al., 2022; Khot et al., 2022; Press et al., 2022; He et al., 2022). Additionally, an iterative retrieval-generation collaborative framework can also be used to refine responses in each iteration based on newly retrieved contexts, following recent advancements in iterative retrieval and generation frameworks for complex tasks (Shao et al., 2023; Feng et al., 2023).

Ethical Considerations

This work aims to provide insights about the automatic generation of model cards and data cards. Such an endeavor is instrumental in promoting accountability and traceability among developers as they document their models. The dataset for this research was collected using public REST APIs from HF Hub, Arxiv, and GitHub. We ensured that only open-source model cards, data cards, and their associated source documents were collected, strictly adhering to the stipulations of their respective licenses for research purposes, so there were no user privacy concerns in the dataset. Our dataset and method should only be used for research purpose.

On the other hand, while the questions we pose to LLMs are technical and specific, there remains a risk of receiving biased responses, particularly for certain queries. For instance, the question about model limitations might yield biased answers, as source papers and GitHub READMEs could contain overstated claims about their models. Consequently, our generated model cards could contain these statements as well if the source texts contain-

ing them are retrieved.

To mitigate this, one reasonable approach is to insert a step after retrieval to filter out or neutralize overstatements. Additionally, we can explicitly prompt LLMs to account for such biases during the generation stage. Another concern is the potential for content homogeneity when using LLMs for model card generation. Excessive reliance on templates could limit model card creators' potential to discuss new issues not covered in the original papers or GitHub repositories (Nakadai et al., 2023; Acion et al., 2023).

Moreover, one aspect of our approach is that we use direct prompts to LLMs rather than fine-tuning them on human-generated model cards, which can also exhibit biases from the internal of LLMs, such as overstatements on well-known models or omissions of potential risks. In our analysis of 2495 human-written model cards in our dataset, only 30.54% mention "weakness(es)" or "limitation(s)", and 15.23% mention "bias(es)". If future study can collect more fairly-written human-generated model cards, they can also be used to finetune LLMs for better performance on this task.

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Wei Zhao, Maxime Peyrard, Fei Liu, Yang Gao, Christian M. Meyer, and Steffen Eger. 2019. [MoverScore: Text generation evaluating with contextualized embeddings and earth mover distance](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 563–578, Hong Kong, China. Association for Computational Linguistics.

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A Question Templates

Tables 10 and 11 shows full question templates of model cards and data cards. We have 31 questions in total for generating model cards, and 21 questions for generating data cards. We create these questions based on the template provided by HF,⁸ and include necessary requirements.

B Dataset Collection Details

For the model card evaluation set selection, we select all 350 examples that are rewritten by the HF team with their unique disclaimers, as shown in Figure 3.

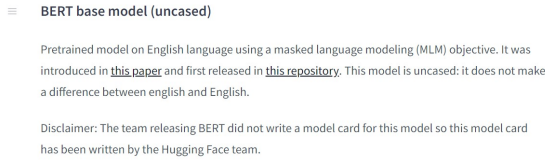


Figure 3: bert-base-uncased (Devlin et al., 2018) as a current model card example with a unique disclaimer sentence, indicating a modification by the HF team.

C Dataset Annotation Details

Human Annotation Guidelines To evaluate paper links and direct GitHub links on the model card evaluation set, we require the annotators to go through each current model card and provide all possible paper links and GitHub links to annotators. They are asked to select the direct paper link and GitHub link from all candidate links, by looking at their positions of occurrences in the model card example. If no direct links of either sources can be determined, they need to label this model card as “Invalid”.

GPT Annotation Details We show our two-shot prompts for asking GPT-3.5-turbo to select direct paper links in Figure 4. Direct GitHub link selection is prompted similarly.

⁸https://github.com/huggingface/huggingface_hub/tree/main/src/huggingface_hub/templates

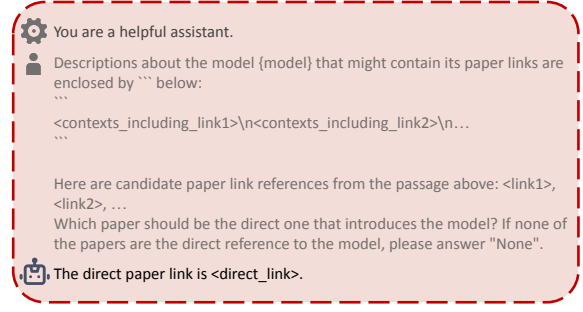


Figure 4: Prompts for calling GPT3.5 to select direct paper links. We prepend one positive example and one negative example to the message list to improve its inference quality.

LLM	# words	# sentences	# links
GPT3.5	4023.88	215.17	4.18
Llama2 70B Chat	6210.32	323.56	4.55
Llama2 7B Chat	5548.50	302.73	1.44
Mistral 7B Inst	4126.07	202.16	2.65

Table 12: Statistics about whole generated model cards

D Dataset Analysis

We provide the number of card examples with direct paper links in their human-generated cards, with direct GitHub repository links, and with both links in Table 13. We also provide additional figures about the dataset task taxonomy in ???. The taxonomy is obtained using the REST API of HF Hub.

	Split	Measure	# w/ papers	# w/ repos	# w/ both
ModelCard	all	# samples	5689	4829	2485
		# words	1064	948	1134
	test	# samples	344	299	294
		# words	668	710	711
DataCard	all	# samples	660	533	328
		# words	1394	1104	1416
	test	# samples	86	71	50
		# words	1003	1290	1155

Table 13: Statistics for crawled model cards and data cards, including the number of examples with direct paper links or direct github links or both, and the average number of words in each category.

E Retriever Details

We use FAISS as our embedding store database (Johnson et al., 2019). We fix the chunk size as 512 and the chunk overlap as 64. After retrieving relevant sections, we choose to obtain 8 chunks from these sections, together with 4 other chunks from other sections to reduce the bias propagation.

Question	Role	Prompt
Summary	Project organizer	Provide a 1-2 sentence summary of what the model is.
Description	Project organizer	Provide basic details about the model. This includes the model architecture, training procedures, parameters, and important disclaimers.
Funded by	Project organizer	List the people or organizations that fund this project of the model.
Shared by	Developer	Who are the contributors that made the model available online as a GitHub repo?
Model type	Project organizer	Summarize the type of the model in terms of the training method, machine learning type, and modality in one sentence.
Language	Project organizer	Summarize what natural human language the model uses or processes in one sentence.
License	Project organizer	Provide the name and link to the license being used for the model.
Finetuned from	Project organizer	If the model is fine-tuned from another model, provide the name and link to that base model.
Demo sources	Project organizer	Provide the link to the demo of the model.
Direct use	Project organizer	Explain how the model can be used without fine-tuning, post-processing, or plugging into a pipeline. Provide a code snippet if necessary
Downstream use	Project organizer	Explain how this model can be used when fine-tuned for a task or when plugged into a larger ecosystem or app. Provide a code snippet if necessary
Out of scope use	Sociotechnic	How the model may foreseeably be misused and address what users ought not do with the model.
Bias risks limitations	Sociotechnic	What are the known or foreseeable issues stemming from this model? These include foreseeable harms, misunderstandings, and technical and sociotechnical limitations.
Bias recommendations	Sociotechnic	What are recommendations with respect to the foreseeable issues about the model?
Training data	Developer	Write 1-2 sentences on what the training data of the model is. Links to documentation related to data pre-processing or additional filtering may go here as well as in More Information.
Preprocessing	Developer	Provide detail tokenization, resizing/rewriting (depending on the modality), etc. about the preprocessing for the data of the model.
Training regime	Developer	Provide detail training hyperparameters when training the model.
Speeds sizes times	Developer	Provide detail throughput, start or end time, checkpoint sizes, etc. about the model.
Testing data	Developer	Provide benchmarks or datasets that the model evaluates on.
Testing factors	Sociotechnic	What are the foreseeable characteristics that will influence how the model behaves? This includes domain and context, as well as population subgroups. Evaluation should ideally be disaggregated across factors in order to uncover disparities in performance.
Testing metrics	Developer	What metrics will be used for evaluation in light of tradeoffs between different errors about the model?
Results	Developer	Provide evaluation results of the model based on the Factors and Metrics.
Results summary	Developer	Summarize the evaluation results about the model.
Model examination	Developer	This is an experimental section some developers are beginning to add, where work on explainability/interpretability may go about the model.
Hardware	Developer	Provide the hardware type that the model is trained on.
Software	Developer	Provide the software type that the model is trained on.
Hours used	Developer	Provide the amount of time used to train the model.
Cloud provider	Developer	Provide the cloud provider that the model is trained on.
Co2 emitted	Developer	Provide the amount of carbon emitted when training the model.
Model specs	Developer	Provide the model architecture and objective about the model.
Compute infrastructure	Developer	Provide the compute infrastructure about the model.

Table 10: Template of the all questions necessary for generating a whole model card.

F Generator Details

Open-sourced LLMs are inferenced through vllm [Kwon et al. \(2023\)](#). Llama2-70B-Chat-HF is run on 4 A6000s. Two 7B models are run on 1 A6000. We fix temperature to 0 to ensure a stable generation quality. We show our prompt description of different roles in Table 14, and the generation prompt in Figure 8.

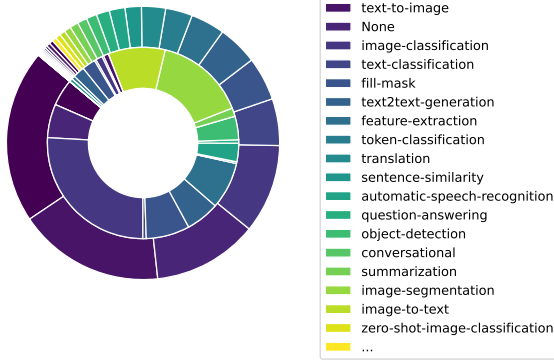
G LLM Generated Model Card Statistics

Statistics about LLM generated model cards are shown in Tables 12 and 16 to 18.

H Metric Details

For standard metrics, we use the list of retrieved texts together with the generated answer as inputs. We normalize all these scores to be in the [0,1] range. Since the output of

Task taxonomy for models in model cards



Task taxonomy for datasets in dataset cards

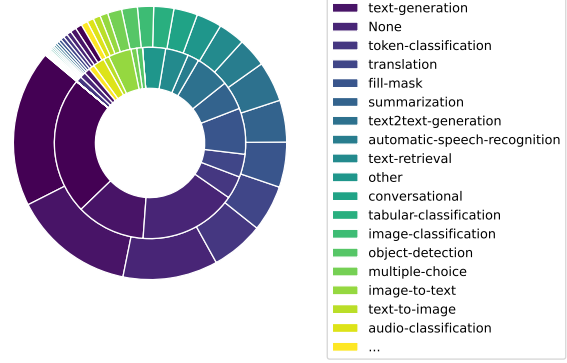


Figure 5: The task taxonomy of models in the model cards dataset (left), and the task taxonomy of datasets in the dataset cards dataset (right), with the inner circle as the test set, and the outer circle as the whole set.

Distribution in Model/Dataset Card Downloads

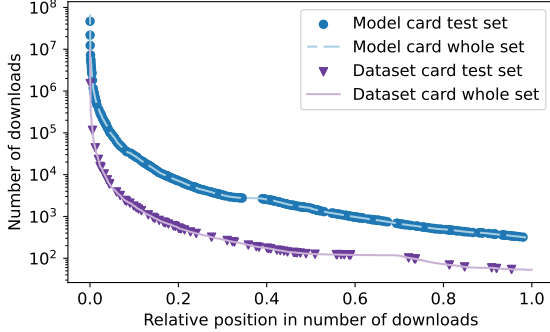


Figure 6: Distribution of the amount of downloads for the whole dataset and the test set. Test set examples distribute quite uniformly.

nli-deberta-v3-large is in {"contradiction", "entailment", "neutral"}, we map these outputs to {0, 0.5, 1}, respectively to maintain a percentage scale. We use the implementation of ROUGE score by HF. We use official implementations for BERTScore and BARTScore.

For GPT metrics, we use GPT-4-1106-preview as evaluators for the main results, and use GPT-3.5-turbo for ablation studies.

I Human Annotation Details

We give two annotators the same set of examples each with seven model cards generated by LLMs and one written by human. For each model example for which LLMs generate model cards, we provide annotators with the model name, the corresponding paper link, GitHub link, and a collection of model cards created by humans or LLMs, as illustrated in Figure 9. We also provide the question template set in Table 10, along with the following

Comparison of Pseudo Answer Length Distributions

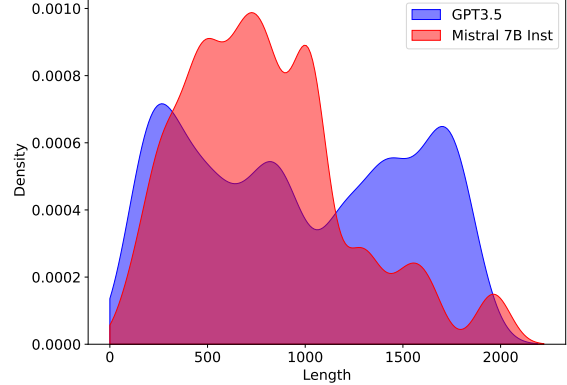


Figure 7: Distribution comparison of pseudo answer length generated by GPT3.5 and Mistral 7B Instruct.

instructions:

Annotators are asked to rank the model cards based on five criteria: completeness, accuracy, objectivity, understandability, and reference quality. The ranking is asked to consider the summation of the binary classification score of whether each question from the model card’s question template is satisfactorily answered according to the specific metric. The final score reported in Table 4 is calculated by simply subtracting the rank from (1 + the total number of candidates). Further, we define each metric as follows:

- **Completeness:** Does the model card comprehensively cover essential aspects such as model summary, description, intended uses, evaluation results, and information about biases or limitations?
- **Accuracy:** Are answers to all the questions in the model card consistent and accurate com-

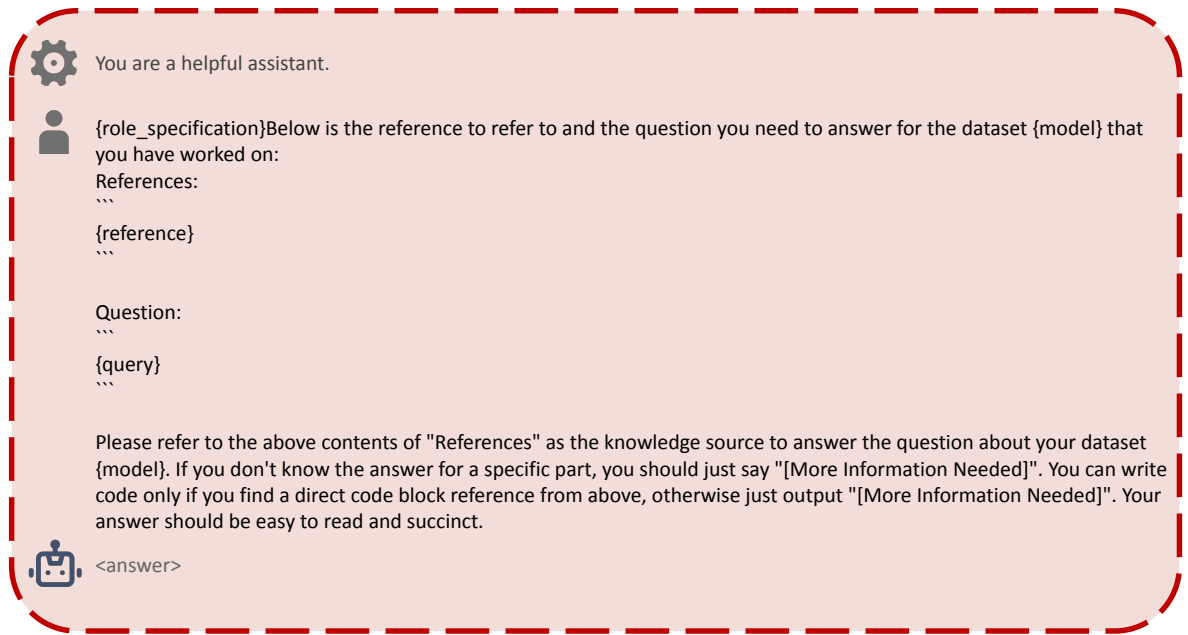


Figure 8: Our generation prompt templates.

pared to the details provided in the model’s official paper and GitHub READMEs?

- **Objectivity:** Does the model card present a balanced perspective of the model, recognizing both its strengths and weaknesses?
- **Understandability:** Is the information in the model card clear and easily understandable for both technical and non-technical audiences? Are complex technical concepts explained in a manner that can be easily grasped by users without in-depth technical knowledge?
- **Reference Quality:** Does the model card include necessary citations and references to related papers and links? Do all provided links redirect correctly to their intended URLs?

In cases where the summation scores for a question are tied for multiple models, we allow annotators the discretion to rank based on the quality of answers to the most important questions, including model summary, description, intended uses, evaluation results, and information about biases or limitations.

We calculate the Krippendorff’s α among the results of two annotators, and got mean=0.68 and std=0.29 for the agreement level. We report averaged ranking scores in Table 4. Note that we don’t have direct comparison across human evaluation metrics vs. automatic metrics, since our human metrics evaluate on a whole model card,

while automatic metrics take each (Q, R, A) tuple for evaluation and they have different scales. We need to implement human metrics in this way to supplement the limited scope of automatic metrics’ focus.

J Retrieval Only Baseline Details

Following [Fu et al. \(2023\)](#), we prompt GPT3.5 to assess the understandability and brevity of generated texts according to input questions. Since there are only two methods to evaluate: retrieval-only and CARDGEN, we use the comparative assessment proposed by [Liusie et al. \(2024\)](#), to compare these two candidates in a pairwise manner. The definition of understandability is the same as in the human annotation. Figures 10 and 11 shows the prompt templates we use to generate comparative results. The order of two candidates in the prompt is randomly shuffled to avoid positional bias. We report the ratio of one method better than another for understandability in Table 8. Full results including brevity are reported in Table 15.

K Pseudo Answer Ablation Study Analyses

We show the distribution of pseudo answer length generated by GPT3.5 and Mistral 7B Instruct in Figure 7.

Model Information

Model: bert-large-cased-whole-word-masking
Paper link: <https://arxiv.org/pdf/1810.04805.pdf>
GitHub link: <https://github.com/google-research/bert>

Model 1's generation

Model Card for bert-large-cased-whole-word-masking

bert-large-cased-whole-word-masking is a pre-trained BERT model that uses whole word masking, where all of the tokens corresponding to a word are masked at once during pre-training. It is a large model with 24 layers, 1024 hidden units, 16 attention heads, and 340M parameters, and is cased, meaning it is case-sensitive.

Model Details

Model Description

Model Card for bert-large-cased-whole-word-masking

Model Details:
- Architecture: 24-layer Transformer encoder, bidirectional self-attention
- Pre-training Objective: Masked Language Model (MLM), predicting masked tokens based on bidirectional context
- Masking Approach: Whole Word Masking (WWM),

Model 2's generation

Model Card for bert-large-cased-whole-word-masking

The model bert-large-cased-whole-word-masking is a pre-trained language representation model developed by the BERT team. It is a variant of the BERT model that uses a new technique called Whole Word Masking, where all the tokens corresponding to a word are masked at once, rather than randomly selecting a few tokens as in the original BERT model. This technique is designed to improve the model's performance on tasks that are too easy for words that have been split into multiple WordPieces. The model is available in both uncased and cased variants, and has 24 layers, 1024 hidden units, and 16 heads. It has 340M parameters in total.

Model Details

Model Description

Here is the information you requested for the BERT-Large-Cased-Whole-Word-Masking model:

Model 3's generation

Model Card for bert-large-cased-whole-word-masking

The model bert-large-cased-whole-word-masking is a pre-trained language representation model that uses a masked language modeling objective to learn bidirectional representations of words in a sentence. It is a variant of the BERT model, with a larger number of layers and hidden size, and is specifically designed for whole word masking, where the model predicts the original vocabulary id of the masked word based only on its context.

Model Details

Model Description

The bert-large-cased-whole-word-masking model is a variant of the BERT-Large model, with an additional technique called Whole Word Masking. In this model, all tokens corresponding to a word are masked at once, rather than just a single token. The training process and architecture remain the same as the original BERT-Large

Model 4's generation

Model Card for bert-large-cased-whole-word-masking

The BERT-Large, Cased (Whole Word Masking) model is a 24-layer Transformer with 1024 hidden units per layer, 16 attention heads, and a total of 340 million parameters, pre-trained on a large corpus including Wikipedia and BookCorpus. It uses whole word masking during pre-training, where all of the tokens corresponding to a word are masked at once, allowing the model to better understand the context of each word in a bidirectional manner.

Model Details

Model Description

Model Card for BERT-Large, Cased (Whole Word Masking)

Model Architecture
BERT-Large, Cased (Whole Word Masking) is a deep learning model based on the Transformer architecture.

Model 5's generation

...
language: en
license: apache-2.0
datasets:
- bookcorpus
- wikipedia
...

BERT large model (cased) whole word masking

Pretrained model on English language using a masked language modeling (MLM) objective. It was introduced in [this paper](https://arxiv.org/abs/1810.04805) and first released in [this repository](https://github.com/google-research/bert). This model is cased: it makes a difference between english and English.

Differently to other BERT models, this model was trained with a new technique: Whole Word Masking. In this case, all of the tokens corresponding to a word are masked at once. The overall masking rate remains the

Model 6's generation

Model Card for bert-large-cased-whole-word-masking

BERT-Large-Cased-Whole-Word-Masking is a pre-trained language representation model that uses Whole Word Masking technique to pre-train deep bidirectional representations from unlabeled text. The model has 24 layers, 1024 hidden units, and 16 heads, with 340M parameters. It can be fine-tuned for a wide range of NLP tasks without substantial task-specific architecture modifications.

Model Details

Model Description

Model Name: BERT-Large-Cased-Whole-Word-Masking

Description:
BERT-Large-Cased-Whole-Word-Masking is a pre-trained bidirectional language representation model that uses a

Model 7's generation

Model Card for bert-large-cased-whole-word-masking

The model "bert-large-cased-whole-word-masking" is a variant of the BERT model that uses whole word masking during training. It is a deep learning model that pre-trains language representations from unlabeled text by jointly conditioning on both left and right context in all layers. The model can be fine-tuned with just one additional output layer to create state-of-the-art models for various NLP tasks. The model was introduced in the paper "Bidirectional Encoder Representations from Transformers" and has been shown to outperform previous methods due to its unsupervised and deeply bidirectional nature.

Model Details

Model Description

Model Card: bert-large-cased-whole-word-masking

Introduction:

Model 8's generation

Model Card for bert-large-cased-whole-word-masking

The model bert-large-cased-whole-word-masking is a 24-layer, 1024-hidden, and 16-heads pre-trained language representation model that utilizes bidirectional pre-training with masked language models to enable deep bidirectional representations, achieving state-of-the-art performance on various sentence-level and token-level natural language processing tasks. It is part of the BERT (Bidirectional Encoder Representations from Transformers) framework, and the pre-trained models can be accessed from the provided link.

Model Details

Model Description

Model Name: bert-large-cased-whole-word-masking

Description:
BERT-Large is a language representation model that is designed to pretrain deep bidirectional representations

Question

Evaluate and rank the model cards generated by GPT4, GPT-3.5, Claude3 Opus, Llama2 70B, Vicuna 13B, Llama2 7B, Mistral 7B, and human on each of the following criteria. Assign a rank score from 1 to 8, where 1 is the highest rank.
1. Completeness: Does the model card comprehensively cover essential aspects such as model summary, description, intended uses, evaluation results, and information about biases or limitations?

Your Rankings

Enter your rankings here in the format like 6,5,4,3,7,8,2,1

Output, just showing what you entered in the "Your Rankings" box and verify that it's correct

The question index you want to go to, only input a value, then enter the goto button

Previous

Submit

Next

GoTo

Figure 9: The human annotation interface built by gradio with an example of model bert-large-cased-whole-word-masking (Abid et al., 2019; Devlin et al., 2018). The information that a model card is written by whom is hidden, and orders of five model cards shown at each time are randomly shuffled to avoid positional bias.

L Generation Only Ablation Study Analyses

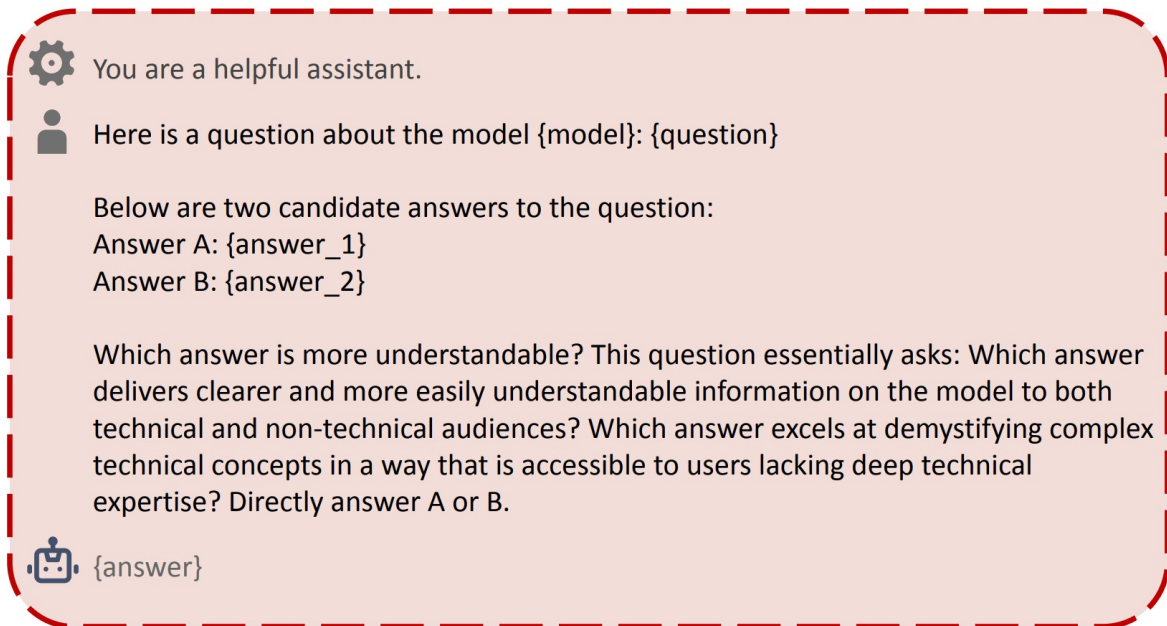


Figure 10: Prompt template to compare CARDGEN's understandability to the retrieval-only baseline.

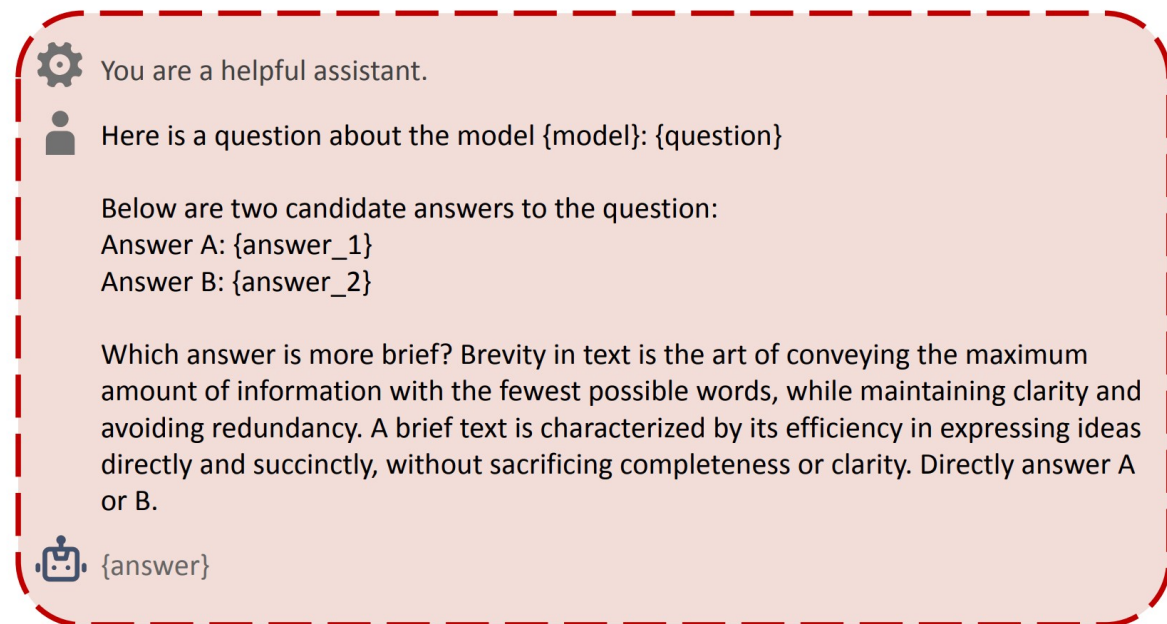


Figure 11: Prompt template to compare CARDGEN's understandability to the retrieval-only baseline.

Question	Role	Prompt
Description	Data manager	Provide the homepage link for the dataset, just give me a link please.
Leaderboard	Data manager	Provide the Leaderboard link for the dataset.
Pointofcontact	Data manager	Provide the Point of Contact for the dataset.
Summary	Data manager	Provide basic details about the dataset. Briefly summarize the dataset, its intended use and the supported tasks. Give an overview of how and why the dataset was created. The summary should explicitly describe the domain, topic, or genre covered.
Supported tasks and leaderboards	Data analyst	Describe the tasks and leaderboards supported by the dataset. Include task description, metrics, suggested models, and leaderboard details.
Languages	Data analyst	Provide an overview of the languages represented in the dataset, including details like language type, script, and region. Include BCP-47 codes if available.
Data instances	Data scientist	Provide a JSON-formatted example of a typical instance in the dataset with a brief description. Include a link to more examples if available. Describe any relationships between data points.
Data fields	Data architect	List and describe the fields in the dataset, including their data type, usage in tasks, and attributes like span indices. Mention if the dataset contains example IDs and their inherent meaning.
Data splits	Data manager	Describe the data splits in the dataset. Include details such as the number of splits, any criteria used for splitting the data, differences between the splits, and the sizes of each split. Provide descriptive statistics for the features where appropriate, for example, average sentence length for each split.
Curation rationale	Data manager	What need or purpose motivated the creation of this dataset? Describe the underlying reasons and major choices involved in its assembly. Explain the significance of the dataset in its field and any specific gaps or demands it aims to address.
Source data	Data manager	Describe the source data used for this dataset. Describe the data collection process. Describe any criteria for data selection or filtering. List any key words or search terms used. If possible, include runtime information for the collection process.
Source language producers	Data manager	Clarifying the human or machine origin of the dataset. Avoiding assumptions about the identity or demographics of the data creators. Providing information about the people represented in the data, with references where applicable.
Annotations	Data manager	Describe the annotation process to the dataset. Detail the annotation process and tools used, or note if none were applied. Specify the volume of data annotated.
Annotators	Data manager	Describe the annotator of the dataset. For annotations in the dataset, state their human or machine-generated nature. Describe the creators of the annotations, their selection process, and any self-reported demographic information.
Personal and sensitive information	Data manager	Categorize how identity data, such as gender referencing Larson (2017), is sourced and used in the dataset. Indicate if the data includes sensitive information or can identify individuals. Describe any anonymization methods applied.
Social impact of dataset	Data manager	Explore the dataset’s social impacts: its role in advancing technology and enhancing quality of life. Consider negative effects like decision-making opacity and reinforcing biases. Check if it includes low-resource or under-represented languages. Assess its impact on underserved communities.
Discussion of biases	Data manager	When constructing datasets, especially those including text-based content like Wikipedia articles, biases may be present. If there have been analyses to quantify these biases, it’s important to summarize these studies and note any measures taken to mitigate the biases.
Other known limitation	Data analyst	Outline and cite any known limitations of the dataset, such as annotation artifacts, in your studies.
Dataset curators	Data manager	List the people involved in collecting the dataset and their affiliations. If known, include information about funding sources for the dataset. This should encompass individuals, organizations, and any collaborative efforts involved in the dataset creation.
Licensing information	Legal advisor	Provide the license and link to the license webpage if available for the dataset.
Contributions	Data manager	Write in 1-2 sentence about the contributors for the dataset. Mention the GitHub username and provide their GitHub profile link. You should follow the format: Thanks to [@github-username](https://github.com/<github-username>) for adding this dataset.

Table 11: Template of the all questions necessary for generating a whole data card.

Card	Role	Description
ModelCard	Developer	who writes the code and runs training
	Sociotechnic	who is skilled at analyzing the interaction of technology and society long-term (this includes lawyers, ethicists, sociologists, or rights advocates)
	Project organizer	who understands the overall scope and reach of the model and can roughly fill out each part of the card, and who serves as a contact person for model card updates
DataCard	Data curator	who collects and organizes the data
	Data analyst	who is skilled at understanding and documenting dataset characteristics and biases
	Data manager	who oversees dataset versioning, availability, and usage guidelines

Table 14: Our prompts for different roles in answering specific questions.

Model	Method	# Words	AR	Understandability	Brevity
GPT3.5	Retrieval only	613.95	81.28	5.60%	1.60%
	CARDGEN	200.74	90.84(+9.56)	94.40%	98.40%
Llama2 70B	Retrieval only	645.91	81.61	1.60%	3.20%
	CARDGEN	230.42	90.32(+8.71)	98.40%	96.80%
Llama2 7B	Retrieval only	603.45	81.32	4.40%	2.80%
	CARDGEN	203.70	90.78(+9.46)	95.60%	97.20%
Mistral 7B	Retrieval only	590.35	81.49	2.40%	2.40%
	CARDGEN	189.11	89.83(+8.34)	97.60%	97.60%

Table 15: GPT3.5 evaluation results of the retrieval-only baseline and CARDGEN on word numbers, answer relevance, understandability, brevity.

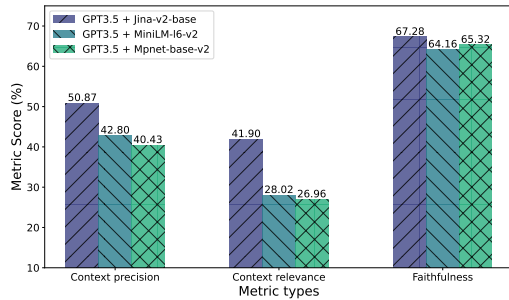


Figure 12: Comparison of three embedding models on context precision, context relevance, and faithfulness.

Question	GPT3.5	Llama2 70B	Llama2 7B	Mistral 7B
Summary	1.95	3.23	2.61	2.39
Description	14.51	13.87	8.93	12.87
Funded by	4.25	4.96	6.40	1.89
Shared by	1.86	4.53	3.18	2.41
Model type	1.51	3.47	2.68	1.70
Language	1.10	4.30	1.84	1.09
License	2.78	4.74	2.79	2.49
Finetuned from	4.81	5.96	5.98	3.47
Demo sources	3.83	7.42	12.81	6.48
Direct use	8.78	7.45	12.20	6.29
Downstream use	10.23	8.11	16.29	7.30
Out of scope use	16.50	21.69	20.71	10.20
Bias risks limitations	19.07	22.76	19.05	16.36
Bias recommendations	18.04	22.13	19.88	18.44
Training data	3.14	4.54	3.31	4.01
Preprocessing	11.06	18.20	13.34	12.46
Training regime	4.82	12.66	7.19	11.08
Speeds sizes times	8.41	12.74	10.62	9.40
Testing data	7.98	9.00	5.55	4.96
Testing factors	13.26	17.23	21.64	11.60
Testing metrics	3.67	14.11	14.20	7.12
Results	7.69	16.85	16.22	10.50
Results summary	9.01	10.94	9.79	6.21
Model examination	11.32	17.74	15.67	8.47
Hardware	1.73	4.29	3.43	1.39
Software	3.50	4.54	2.45	1.47
Hours used	2.06	7.52	8.29	2.86
Cloud provider	1.82	4.38	2.92	1.32
Co2 emitted	2.40	9.14	10.27	2.13
Model specs	10.52	12.12	9.90	7.17
Compute infrastructure	3.59	12.94	12.61	6.61

Table 17: Number of sentences in generated model cards per question averaged by all samples in the test set.

Question	GPT3.5	Llama2 70B	Llama2 7B	Mistral 7B
Summary	53.91	89.40	71.93	63.61
Description	275.47	276.50	187.40	264.11
Funded by	78.29	96.10	91.97	31.15
Shared by	33.41	108.62	57.94	43.69
Model type	46.11	115.77	67.69	56.07
Language	30.24	100.23	57.52	21.67
License	47.56	94.86	43.05	42.63
Finetuned from	93.95	137.65	115.16	65.91
Demo sources	76.70	150.54	228.09	141.35
Direct use	227.26	247.95	260.14	211.97
Downstream use	287.05	256.03	301.56	254.17
Out of scope use	305.64	341.98	339.81	225.52
Bias risks limitations	305.09	330.94	317.83	274.26
Bias recommendations	298.46	333.96	336.44	309.82
Training data	61.17	103.41	72.18	85.98
Preprocessing	169.67	285.66	228.65	222.67
Training regime	110.71	208.14	162.46	179.76
Speeds sizes times	170.33	250.69	211.52	192.81
Testing data	112.20	144.15	87.29	87.16
Testing factors	230.03	293.02	344.08	245.14
Testing metrics	64.45	267.89	226.08	137.77
Results	137.94	276.72	263.82	210.40
Results summary	154.57	230.82	215.33	136.51
Model examination	214.29	317.01	264.26	169.52
Hardware	24.87	81.48	72.26	21.44
Software	64.71	91.29	49.32	23.53
Hours used	27.95	172.74	164.28	58.86
Cloud provider	26.13	82.82	56.88	18.55
Co2 emitted	36.01	220.29	243.23	33.65
Model specs	207.91	276.66	204.47	161.47
Compute infrastructure	51.80	227.01	205.86	134.92

Table 16: Number of words in generated model cards per question averaged by all samples in the test set.

Question	GPT3.5	Llama2 70B	Llama2 7B	Mistral 7B
Summary	0.02	0.05	0.00	0.01
Description	0.17	0.04	0.01	0.04
Funded by	0.37	0.06	0.05	0.06
Shared by	0.36	0.58	0.04	0.12
Model type	0.00	0.00	0.00	0.00
Language	0.01	0.00	0.00	0.01
License	0.53	0.82	0.17	0.36
Finetuned from	0.26	1.06	0.30	0.49
Demo sources	0.66	0.82	0.51	0.94
Direct use	0.34	0.05	0.01	0.09
Downstream use	0.17	0.03	0.02	0.04
Out of scope use	0.20	0.00	0.00	0.00
Bias risks limitations	0.01	0.00	0.00	0.00
Bias recommendations	0.04	0.01	0.00	0.00
Training data	0.29	0.24	0.00	0.02
Preprocessing	0.04	0.03	0.00	0.01
Training regime	0.00	0.03	0.01	0.01
Speeds sizes times	0.21	0.10	0.02	0.05
Testing data	0.01	0.01	0.03	0.02
Testing factors	0.01	0.00	0.00	0.01
Testing metrics	0.01	0.02	0.00	0.01
Results	0.03	0.05	0.03	0.04
Results summary	0.04	0.03	0.04	0.09
Model examination	0.19	0.04	0.02	0.02
Hardware	0.00	0.02	0.04	0.01
Software	0.03	0.12	0.00	0.04
Hours used	0.01	0.02	0.00	0.01
Cloud provider	0.03	0.11	0.04	0.02
Co2 emitted	0.01	0.11	0.00	0.00
Model specs	0.11	0.04	0.01	0.03
Compute infrastructure	0.02	0.05	0.05	0.10

Table 18: Number of links in generated model cards per question averaged by all samples in the test set.